

11 Cell Communication

KEY CONCEPTS

- 11.1** External signals are converted to responses within the cell *p. 213*
- 11.2** Signal reception: A signaling molecule binds to a receptor, causing it to change shape *p. 217*
- 11.3** Signal transduction: Cascades of molecular interactions transmit signals from receptors to relay molecules in the cell *p. 221*
- 11.4** Cellular response: Cell signaling leads to regulation of transcription or cytoplasmic activities *p. 226*
- 11.5** Apoptosis requires integration of multiple cell-signaling pathways *p. 229*

Study Tip

Make a table: As you read about examples of cell signaling in this chapter, make a table and classify the events of each example into three stages: signal reception, signal transduction, and cellular response.

Example of Cell Signaling	Signal Reception	Signal Transduction	Cellular Response
Epinephrine	Epinephrine binds cell-surface receptor.	Relay molecules each activate the next molecule.	An enzyme is activated that breaks down glycogen into glucose for energy to fight or flee.

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For Students (in eText and Study Area)

- Get Ready for Chapter 11
- Animation: Overview of Cell Signaling
- Animation: Signal Transduction Pathways

For Instructors to Assign (in Item Library)

- Problem-Solving Exercise: Can a Skin Wound Turn Deadly?
- Tutorial: Cell-Signaling: Transduction and Response



Figure 11.1 This impala is fleeing for its life, racing to escape the predatory cheetah nipping at its heels. The impala is breathing rapidly, its heart pounding and its legs pumping furiously. These physiological functions are all part of the impala's "fight-or-flight" response, driven by hormones released from its adrenal glands at times of stress—in this case, upon sensing the cheetah.

How does cell signaling fuel the desperate flight of an impala?

